By substituting $f(-x)=-f(x)$ for $f(-x)=f(x)$, we obtain the definition of an $\boxed{\text{odd}}$ function.

But let us return to monotonic functions.

If we drop the equality sign in the definition of a monotonic function, (in $f(x_1)\leqslant f(x_2)$ or $f(x_1)\geqslant f(x_2)$), we obtain a so-called $\boxed{\text{strictly monotonic function}}$. In this case a nondecreasing function becomes an $\boxed{\text{increasing function}}$ (i.e. $f(x_1)<f(x_2)$). Similarly, a nonincreasing function becomes a $\boxed{\text{decreasing function}}$ (i.e. $f(x_1)>f(x_2)$). In all the previous illustrations of monotonic functions, we actually dealt with strictly monotonic functions (either increasing or decreasing).

∘ Strictly monotonic functions possess an interesting property: $\boxed{\text{each has an inverse function}}$

$\boxed{\text{Reader:}}$ The concept of an inverse function has already been used in the prev dialogue in conjunction with the possibility of mapping a set of equilateral triangles onto a set of circles. We saw that the $\boxed{\text{inverse mapping}}$, i.e. the mapping of the set of circles onto the set of equilateral triangles, was possible.

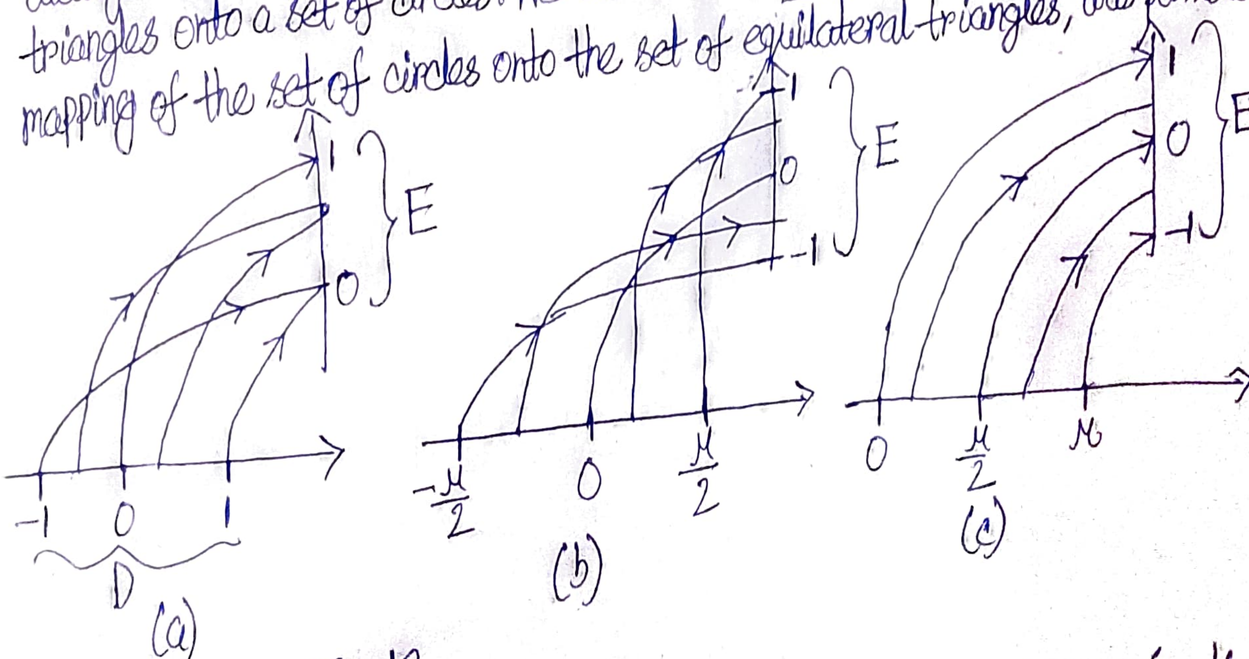

Fig. 18

$\boxed{\text{Author:}}$ That's right. Here we shall examine the concept of an inverse function in greater detail (but for numerical functions). Consider Fig. 18. Similarly to the graphs presented in Fig. 13, it shows three functions: (a) $y=\sqrt{1-x^2}$, $-1\leqslant x\leqslant 1$

(b) $y=\sin x$, $-\frac{\pi}{2}\leqslant x\leqslant\frac{\pi}{2}$ (c) $y=\cos x$ $0\leqslant x\leqslant\pi$